

HIT137

Group Assignment 3 (30% Mark)

You are required to create a GitHub repository and add all your group mates to it (make sure to keep it public, not private). You should do this before you start the assignment.

All the answers and contributions should be recorded in GitHub till you submit the assignment.

Submission Guidelines:

- **Include your GitHub Repository link in a text file “github_link.txt”.**
- **Zip all the programming files, outputs, and “github_link.txt”. Upload them to Learline.**

Late submissions: A late submission penalty of 5% of the total available marks per day will apply.

Overview

You will develop a desktop application that demonstrates your understanding of OOP, GUI development using Tkinter, and image processing using OpenCV.

Build a desktop application where a loaded image is cut into a scrambled grid of tiles. The image starts as a complete picture; it is then programmatically transformed by moving, rotating, and flipping the pieces. The player interacts with the tiles to restore the picture, and the application tracks the player's moves, offers hints, and detects when the picture has been fully restored.

Functional Requirements

1. Object-Oriented Programming

Your application must be structured using proper OOP principles:

- Demonstrate encapsulation, constructor, methods, class interaction, inheritance and polymorphism

2. Image Processing with OpenCV

When an image is loaded, the application should:

- Resize the image so it fits on screen, then crop or pad it so it divides evenly into a grid. The default grid is 3×3 ; the player must be able to choose 3×3 , 4×4 or 5×5 before loading.

- Programmatically alter the tiles using at least three types of transformation, chosen randomly each time an image is loaded. All transformations are generated at once. At minimum you must implement:
 - Swap – two tiles exchange positions.
 - Rotate – a tile is rotated by 90° , 180° or 270° .
 - Flip – a tile is flipped horizontally or vertically.
- The number of transformations applied should scale with the grid size (for example, 6 for 3×3 , 12 for 4×4 , 20 for 5×5).
- Reassemble the transformed tiles into a single image for display.

3. Tkinter GUI

Image Loading and Display

- A button allows the user to choose an image from disk, support for common image formats (JPG, PNG, BMP).
- A control lets the player choose the grid size before loading.
- The original image and the transformed image must be displayed side by side. The original image is shown on the left for visual reference only. The transformed image is shown on the right and is the only image that responds to player clicks.
- A faint grid should be drawn over the transformed image so tile boundaries are visible.

Solving the Puzzle

The player restores the picture by clicking on the transformed tile:

- Left click on a tile selects it, the selected tile is highlighted with a coloured border. Left clicking a second tile swaps the two and clears the selection. Left clicking the same tile again deselects it.
- Right click on a tile rotates it 90° clockwise.
- Shift + left click on a tile flips it horizontally.

Whenever a tile is in its correct position and correct orientation, a small green tick should be drawn in its corner so the player knows that tile is done.

Moves and Score

- A visible section of the interface displays the number of moves used on the current image (each swap, rotate or flip counts as one move) and the number of tiles still incorrect.
- When every tile is in its correct position and orientation, the player must be notified and no further input on the puzzle should be accepted. The player can then load another image to continue playing.

Hints

- A `Hint` button marks one currently incorrect tile with a blue circle on the transformed image and marks the tile's correct home position with a blue circle on the original image. The blue circle disappears after the next move. The player may use at most 3 hints per image; after that the button must be disabled.
- A `Solve` button instantly solves the puzzle by undoing all remaining transformations and clear the moves and score.